AUTOMATIC CLASSIFICATION OF CRIME-RELATED TEXT REPORTS USING NLP

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CHAPTER I: INTRODUCTION

1.1 Background Of The Study

A crime is an illegal act that is penalized by the state or another authority. The term “crime” refers to an offence that harms not only a single person but also a community, society, or state (Khan et al., 2022).

Criminal activity is present daily, in a multiplicity of illegal actions in several domains, such as drug trafficking, computer crime, and theft, to just mention a few examples. Upon the occurrence of a crime, criminal police investigators are in charge and start a set of actions to enable the construction of the so-called 'chain of custody' to identify the alleged culprits and to present them in court(Carnaz et al., 2021).

Classification of Crime Reports is an important process in the analysis and management of crime-related data. In this Context: An ID, such as the type of crime, date and time of the incident, location, witnesses, and description of the incident(Hariguna & Ruangkanjanases, 2023).

Text classification specifically focuses on categorising and organising text data to facilitate easier management and analysis. These techniques leverage natural language processing, machine learning, and data mining to extract meaningful patterns and categorize text data. thereby transforming unstructured text into structured, actionable insights (Taha et al., 2024).

Crime-related data is exploding due to the constant reporting and distribution of crimerelated information on the internet in the rapidly evolving information technology and communication era. Online newspapers and social networks offer real-time information about what is happening in a specific location, at a particular moment, and with numerous causes (Ali et al., 2023).

Natural language processing is an area of computer science that is primarily concerned with transforming natural language into a formal language, such as a programming language, so that a computer can process and understand it, mainly by combining algorithms. (Martínez Hernández et al., 2026)

Automated Crime Report Analysis is a technological approach used to automate the process of evaluating, sorting, and understanding crime reports that enter the police system or relevant government agencies(Hariguna & Ruangkanjanases, 2023).

The rise in cybercrime and the complexity of multilingual and code-mixed complaints present significant challenges for law enforcement and cybersecurity agencies. These organizations need automated, scalable methods to identify crime types, enabling efficient processing and prioritization of large complaint volumes. Manual triaging is inefficient, and traditional machine learning methods fail to capture the semantic and contextual nuances of textual cybercrime complaints(Rani et al., 2024).

The proposed research paper explores the application of machine learning techniques in the crime analysis problem, specifically focusing on the classification of crime-related textual data. Through a comparative analysis of various machine learning models, including traditional approaches and deep learning architectures, the study evaluates the data (Mussiraliyeva & Baispay, 2024)

The 2021 amendment to South Korea's Criminal Procedure Law has significantly enhanced the role of the police as investigative authorities. Consequently, there is a heightened demand for advanced investigative expertise among the police, driven by an increase in the number of cases each investigator handles and the extended time required for report preparation. This situation underscores the necessity for an artificial-intelligence-supported system to augment the efficiency of investigators. (Park et al., 2024).

The ability to be anonymous on social media and the lack of adequate resources to fight cybercrime are catalysts for the rise in criminal activities, especially in South Africa. We proposed a system that will analyse and detect crime in social media posts or messages. The new system can detect attacks and drug-related crime messages, hate speech, and offensive messages(Lombo Lombo et al., 2022).

The integration of machine learning techniques in crime analysis offers unprecedented opportunities to harness the vast amount of online textual data for enhancing public safety and security. By overcoming inherent challenges and leveraging the capabilities of ML algorithms, researchers and law enforcement agencies can gain invaluable insights into criminal behaviour and trends, ultimately contributing to more effective crime prevention and intervention strategies (Mussiraliyeva & Baispay, 2024).

In today's world, security is the most prominent aspect that has been given higher priority. Despite the rapid growth and usage of digital devices, the measurement of crimes in underdeveloped countries is still challenging(Umair et al., 2020)

The majority of Somalia's current crime reporting and management systems are manual. Labor-intensive and prone to human error. By creating an automated NLP framework especially suited for the Somali language, this study seeks to address these issues. It is anticipated that this transition from manual sorting to AI-driven analysis will greatly speed up law enforcement responses and enhance data integrity in the Somali security industry.

1.2 Problem Statement

Crime reports sent using digital channels (such as social media, emails, or online forms) ought to be processed and categorised instantly in a perfect law enforcement setting. This enables prompt prioritisation and routing to the appropriate department (such as cybercrime, homicide, or theft), guaranteeing a prompt response that can save lives and stop additional criminal behaviour.

The majority of crime report classifications are now done by hand. Every day, thousands of text-based reports must be evaluated by human dispatchers or officers to determine the type of crime. Today's massive amount of digital data makes this manual approach labor-intensive, slow, and prone to major delays. The use of manual classification results in several serious problems: Slow Reaction: When a report is waiting in a queue, human error – fatigue brought on by information overload – causes reports to be misclassified or high-priority threats to be missed. Resource inefficiency: Instead of actively investigating and ensuring community safety, skilled staff devote hours to administrative sorting. This study suggests using natural language processing (NLP) to create an automated classification system. The system will be trained to examine the linguistic patterns of crime reports, classify them automatically with high accuracy, and identify urgent instances for prompt attention by utilising machine learning methods. The goal of this shift from manual to AI-driven sorting is to significantly increase the speed of police and court responses.

1.3 Research Objectives

1.3.1 General Objective

The main objective of this research is to develop and put into use an intelligent automated framework that uses machine learning and natural language processing to transform the way Security agencies classify crime reports. This framework will replace manual sorting with high-Speed semantic analysis that guarantees operational efficiency and data integrity.

1.3.2 Specific Objectives

- To reduce the reliance on manual methods of classifying crime reports.
- To collect and preprocess crime-related text data for analysis.
- To extract and analyse key features from crime-related text data for accurate classification.
- To develop an automated NLP system for classifying crime-related text reports.

1.4 Research Questions

1. How to reduce the reliance on manual methods of classifying crime reports?
2. How to collect and preprocess crime-related text data for analysis?
3. How to extract and analyse key features from crime-related text data for accurate Classification?
4. How to develop an automated NLP system for classifying crime-related text reports?

1.5 Significance of the Study

The goal of this project is to create an AI-powered natural language processing system that can automatically categorize text reports pertaining to crimes, making it noteworthy. Researchers and law enforcement organizations can save time and money on manual data analysis by automating the classification process. Decision-making procedures may be improved; the system can help create intelligent crime monitoring tools, and it can increase the accuracy of identifying various sorts of crimes. Additionally, by showing how Natural Language Processing (NLP) It may be used practically in crime analysis; it advances the academic subject of NLP.

1.6 Scope of the Study

The design, development, and assessment of a system based on Natural Language Processing (NLP) that automatically ascertains whether a given text report is related to a crime or not is the main objective of this study.

1.7 Organization of the Study

I. **Chapter II:** Literature Review The Literature Review chapter is the summary of the study conducted by the researcher on the existing literature and studies that are associated with the topic of the study. This study reviews the existing research conducted to determine the effectiveness of crime detection and text classification using Natural Language Processing (NLP) techniques. Additionally, this chapter aims to identify the various machine learning algorithms and approaches that have been implemented on social media data to classify and identify crimes.

II. **Chapter III**: Methodology The third chapter discusses the methodology used for the research in this study. It shows how the social media data is collected and preprocessed. That is, in the stage of data preprocessing, the texts in social media posts are cleaned, tokenized, and the useful features are extracted. This chapter also provides more light about the machine learning algorithms that are employed in this study to classify the social media posts as crime and non-crime. Hence, the researcher fully explained data preparation and selection of the machine learning algorithm used.

III. **Chapter IV** System Analysis, Design, and Implementation. This chapter will discuss the analysis, design, and implementation of the proposed system, which is an NLP system to classify social media posts. It will discuss the system architecture, data flow, text preprocessing, feature extraction, and classification of social media text. It will also show the flow of social media text from the input to the output of the system. It will also discuss the coding structure of the system and the deployment of the system.

IV. **Chapter V:** Conclusion and Recommendations This chapter concludes and summarizes all the findings from the study. It also discussed the validity of the experimental results. The study discussed how the NLP model is used to classify crimes on social media and some possible ways of improving it. Recommendations were made on how to improve the performance of the system. These include increasing the number of samples for training and re-training the model for accuracy. The system could also be integrated into existing policing platforms for effective crime control.

CHAPTER II: LITERATURE REVIEW

2.1Introduction

Crimes are the most common social issues nowadays, affecting the economic growth, quality of life, and economy of any country. Crimes affect a country's reputation on an international scale and its economy by imposing a financial burden on the government to hire additional police forces. For the eradication of crimes, the government needs to adopt an optimized strategy and sustainable e-governance information systems. Algorithms that predict the occurrence of crimes based on time and location can help the government to deploy law enforcement in highly dangerous areas
(Umair et al., 2020)

Criminal activity is present daily, in a multiplicity of illegal actions in several domains, such as drug trafficking, computer crime, and theft, just to mention a few examples. Upon the occurrence of a crime, criminal police investigators are in charge and start a set of actions to enable the construction of the so-called chain of custody to identify the alleged culprits and to present them in court. These actions are multidisciplinary and, depending on the crime, they may involve distinct tasks, such as digital and biological forensics analysis, interviews with witnesses, and interrogations with suspects and other individuals who may be potentially implicated(Carnaz et al., 2021)

The classification of crime reports is a crucial process in the analysis and management of crime-related data. In this context, a crime report is written or digital information that contains various details about the crime incident, such as the type of crime, date and time of the incident, location, witnesses, and description of the incident. The goal of crime report classification is to group these reports into appropriate groups based on their characteristics, which facilitates efficient analysis, monitoring, and action by the authorities. One of the main benefits of classifying crime reports is their ability to aid law enforcement and investigation. By classifying crime reports by crime type, authorities can more easily identify potential crime patterns.(Hariguna & Ruangkanjanases, 2023)

Crime-related data is exploding due to the constant reporting and distribution of crime-related information on the internet in the rapidly evolving information technology and communication era. Online newspapers and social networks offer real-time information about what is happening in a specific location, at a particular moment, and with numerous causes. These facts depict real-world events, such as accidents, innovations, conferences, natural disasters, or crime events. These resources have provided a substantial amount of crime data, which is the domain of their study, presenting a need to extract precise and valuable information to promote and strengthen the fighting capacities against crime capabilities, thus improving public protection and reducing the likelihood of future crime prospects(Ali et al., 2023)

Criminal detection and prediction constitute an essential practice in crime analysis, seeking to provide effective techniques for supporting law enforcement. Researchers develop diverse strategies and solutions utilizing machine learning (ML) and deep learning (DL) algorithms to foresee or detect criminal activities that have demonstrated considerable potential in addressing crime detection issues (Al-harbi & Alghieth, 2025) Natural language processing is an area of computer science that is primarily concerned with transforming natural language into a formal language, such as a programming language, so that a computer can process and understand it, mainly combining techniques from AI, linguistics, statistics, and computing for the development of algorithms. It is a fundamental part of NLP as it focuses on understanding the meaning of words, phrases, or sentences and how they are used in a specific context (Martínez Hernández et al., 2026)

Crimes and accidents can have significant economic repercussions that affect not just individuals but entire towns and countries. Even though there have Despite reports of declining crime rates in many affluent countries, it is still clear that significant crimes and accidents continue to occur often in emerging nations, impeding economic growth and causing fatalities and serious injuries. Law enforcement, legislators, and other relevant parties might benefit from timely monitoring of the course of crimes in their proactive and long-term efforts to prevent and resolve the underlying causes of these unfavorable situations. (Pongpaichet et al., 2024)

Extracting events from news stories as the aim of several Natural Language Processing (NLP) applications (e.g., question answering, news recommendation, news summarization) is not a trivial task, due to the complexity of natural language and the fact that news reporting is characterized by journalistic style and norms. Those aspects entail scattering an event description over several sentences within one document (or more documents), applying a mechanism of gradual specification of event-related information. (Bonisoli et al., 2023)

Automated Crime Report Analysis is a technological approach used to automate the process of evaluating, sorting, and understanding crime reports that enter the police system or relevant government agencies. The main objective of this method is to improve efficiency and effectiveness in crime data management, as well as enabling authorities to respond to crime events faster and more accurately. Automated analysis of crime reports involves using technologies such as artificial intelligence and natural language processing to extract key information from incoming crime reports. (Hariguna & Ruangkanjanases, 2023)

The rise in cybercrime and the complexity of multilingual and code-mixed complaints present significant challenges for law enforcement and cybersecurity agencies. These organizations need automated, scalable methods to identify crime types, enabling efficient processing and prioritization of large complaint volumes. Manual triaging is inefficient, and traditional machine learning methods fail to capture the semantic and contextual nuances of textual cybercrime complaints(Rani et al., 2024)

The proposed research paper explores the application of machine learning techniques in the crime analysis problem, specifically focusing on the classification of crime-related textual data. Through a comparative analysis of various machine learning models, including traditional approaches and deep learning architectures, the study evaluates their effectiveness in accurately detecting and categorizing crime-related text data. The performance of the models is assessed using rigorous evaluation metrics, such as the area under the receiver operating characteristic curve (AUC-ROC), to provide insights into their discriminative power and reliability(Mussiraliyeva & Baispay, 2024)

The 2021 amendment to South Korea's Criminal Procedure Law has significantly enhanced the role of the police as investigative authorities. Consequently, there is a heightened demand for advanced investigative expertise among the police, driven by an increase in the number of cases each investigator handles and the extended time required for report preparation. This situation underscores the necessity for an artificial intelligence-supported system to augment the efficiency of investigators. In response, their study designs a hybrid model that fine-tunes two Transformer-based pretrained language models to extract 18 key pieces of information from legal documents automatically(Park et al., 2024)

The Web is used by criminals for a variety of purposes, including creating virtual communities, mobilizing, providing information, offering online training, spreading their services or beliefs, attracting new members, fundraising, and reducing risks. In actuality, Web material and its communication features offer benefits for organizing and carrying out illegal activities. Conversely, the Web can help with the creation of software tools and systems that facilitate criminal investigations and prevention. Over the past few decades, crime rates have rapidly increased in a number of nations, including Brazil. This makes it more difficult to respond to criminal incidents in a prompt and efficient manner. Because of this, automated services are essential for both the investigative and preventive procedures. (Resende de Mendonça et al., 2020)

The integration of machine learning techniques in crime analysis offers unprecedented opportunities to harness the vast amount of online textual data for enhancing public safety and security. By overcoming inherent challenges and leveraging the capabilities of ML algorithms, researchers and law enforcement agencies can gain invaluable insights into criminal behaviors and trends, ultimately contributing to more effective crime prevention and intervention strategies(Mussiraliyeva & Baispay, 2024)

In the contemporary era, ensuring security has become a critical priority due to the rapid proliferation of digital technologies and devices. As digital adoption increases globally, especially in developing regions, monitoring and accurately addressing crime becomes increasingly complex and challenging. Research suggests that developing countries face persistent cybersecurity challenges and limitations in effectively detecting and combating digital crime and security risks in these environments (Yadav et al., 2020)

Social Network Service (SNS) platforms like X (Twitter), Telegram, Meta Threads, and Reddit have surfaced as a result of the introduction of mobile devices and the quick advancement of Internet technology. This makes it possible for people to freely share their thoughts, generating enormous amounts of data instantly. Users can acquire high-quality information thanks to this data explosion, but they are unavoidably exposed to significant cybercrimes. has been revealed that 43% of American youths have experienced social media abuse. According to a Ditch the Label and Habbo poll of 10,008 teenagers between the ages of 13 and 22, almost 37% of them experienced frequent cybercrimes. (Kim et al., 2024)

The duty court in the city where the criminal act occurred receives criminal complaints (police reports) made by the police or security forces in the Spanish legal system. These reports are then reviewed for potential prosecution. These reports are text documents with a simple, natural-language format. A report is categorized by the type of criminal behavior it describes when delivered to the court. A court and, if desired, a prosecutor perform this classification to identify one of three probable courses of action: (a) it is approved and a trial is set; (b) the investigation should proceed if there is insufficient evidence to hold a trial; (c) the report is submitted if the evidence is extremely poor, such as when the perpetrator of the crime or the stolen item cannot be identified (Navarrete et al., 2025)

2.2 Impact of crimes on social media

Social media representations of crime have the power to influence how the general public views crime, law enforcement, and the legal system. Inaccurate coverage of crime-related news can lead to mistakes, portrayal of crime as a necessity and a demand of the time, which makes people sympathetic to the criminal and, in a way, encourages crime. It also undermines public confidence in the government and gives power to those who have taken the law into their own hands. Hate crime is the overrepresentation of crimes committed

by a certain segment of society against that segment, which has become a social media trademark and fosters hatred and negative perceptions toward that segment.
(Abhishek Kumar, 2025)

The Social media exposure of crime refers to the way crime-related content is shared, discussed, and highlighted on social media platforms. Social media is now a major tool for people to know what is happening around them. It is now a necessary part of everyone's life, even for students. As the use of social media as a platform for gathering information continues, individuals will encounter posts that are related to crime incidents. People will have their own insights based on the topic, which might affect them in some ways. Social media networks make it simple to spread both positive and negative emotions among individuals. Criminology students often recognize that social media can serve as an educational tool. By exposing users to real-time information about crime trends and safety issues, social media can enhance public awareness(Ezekwe, 2025)

According to the data, a large number of students share excessive amounts of information via social media and emails, which could endanger them. The majority of organizations have put in place multiple security measures to prevent information from leaking on social media. Cyberbullying has become widespread, particularly among those who are vulnerable to it. Social media is frequently the target of security threats such as hacking, spoofing, spamming, cyberstalking, Trojan horses, worms, viruses, and exposure scams; for this reason, government and ICT policies have addressed these problems. Higher education institutions must devote more funds to student education, awareness, research and development, IT security, and social media crimes(Benard et al., 2021)

Social media is substantially affecting the world's dynamics and proves to be particularly true when it comes to the legal realm.3 As a matter of fact, in this digital era, these massive online platforms give rise to new challenges and opportunities in relation to the existing framework of international criminal law (hereinafter 'ICL')(Ali, 2021)

The penetration of social media into nearly every aspect of contemporary life has reshaped patterns of communication, information consumption, and interpersonal interaction. Beyond its role as a tool for social connectivity, digital platforms have emerged as influential spaces where narratives of crime circulate rapidly and where offenders increasingly exploit online affordances to facilitate unlawful behavior. Their study investigates the extent to which these platforms affect both criminal conduct and public interpretations of crime trends within modern societies. The purpose of the research is to analyze how social media environments enable planning, performance, and dissemination of criminal acts, while also shaping public emotions, risk perceptions, and trust in institutions responsible for maintaining public order(Rashid et al., 2025)

These days, people utilize a variety of social media platforms to share information with friends, family, and other people for official, business, and personal reasons. Cyberstalking is another major problem brought on by social media use. Cyberstalking has been recognized as an increasingly prevalent anti-social issue that impacts victims, educational institutions, and human society as a whole. Cyberstalking on social media must be detected by an intelligent system(Gautam & Bansal, 2022)

Numerous risks that have been recognized and categorized are present on social media, according to the International Journal of Network Security & Its Applications. Al Hasib categorized social networking threats into three categories: identity-related threats, which include phishing, friending malicious actors, profile squatting, stalking, and corporate espionage; privacy-related threats, which include the sharing of private information on social media; and information security threats, which include well-known security threats on social media like worms, viruses, and cross-site scripting. Similar to this, Fire et al. divided the threats associated with social media (Umeugo, 2023)

In today’s digital age, the widespread growth of the digital media sector and information technology has significantly improved communication flexibility through platforms such as Facebook, Twitter, Instagram, and others. These platforms, driven by Web 2.0 technologies, enable global connectivity, fostering social interaction and community building beyond geographical constraints. Despite their advantages, such as accessibility and interactivity, social media also pose risks like cybercrimes and threats to security, making it crucial to evaluate their impact on societal security in the United Arab Emirates (UAE)(Alnaqbi & Ali, 2025)

For many years, one of the primary subjects covered by the news media has been crime. A survey conducted in the United States found that over 70% of the 100 news stations it looked at began with a report about crime. Many other Western cultures exhibit similar patterns. For instance, the United Kingdom has seen an increase in crime reporting since the conclusion of World War II. Similar trends have been seen in Sweden since the 1960s, and since the late 1970s, academics in Finland have noticed an increase in crime coverage in both main evening television news and tabloid headlines. Therefore, one of the main "selling" points for various media outlets in their quest for readers and viewers is crime, especially violent crime (Näsi et al., 2021)

The goal of the study is to examine the TikTok app's features, applications, and content in order to comprehend how it contributes to crime and deviance in society. Additionally, it clarifies the features of crimes and deviations that originate from TikTok in Bangladesh. The study's methodology consists of case studies of fifteen TikTok-related crime instances, qualitative content analysis of 10 of Bangladesh's top TikTok profiles, and convenience sampling of TikTok's "For you" page with particular hashtags. Performance crime, copycat crime, and fame-driven crime are some of the unique traits of crime and deviance associated with TikTok that the study discovered. The study made an effort to pinpoint particular crimes that have previously been committed with this software, including rape, murder, sexual assault, fraud, etc.(Shejuti, 2023)

Crime analysis and prediction utilizing social media activity is a major area of study nowadays. Their study suggests integrating geolocated Twitter data and a subset of violent tweets as dynamic data for increased predictive accuracy, in contrast to the crime prediction models described in the literature that use historical crime data, demographics, socioeconomic, and built environment data as explanatory variables. The assessment of a violent subset in crime prediction models is new, even though Twitter data based on geolocation or topic extraction has been covered in earlier research. Our research demonstrates how crucial it is to extract useful information from large amounts of data rather than utilizing all of the data without comprehending its complexity(Ristea et al., 2020)

2.3 Traditional Classification techniques in crimes

Traditional methods of crime prevention and detection have played an essential role in maintaining law and order within communities. These methods are characterized by their reliance on human intervention, community participation, and well-established policing practices. This section discusses these key traditional methods, highlighting their strengths and limitations. (Apene et al., 2024)

The efficacy of several machine learning models, such as LSTM, GRU, Random Forest, SVM, and Logistic Regression, is examined in their study. Based on thorough testing, the Random Forest, SVM, and Logistic Regression models performed well on tasks involving the classification of crimes. Traditional inspection of a large number of crime reports, however, may be laborious and prone to human error. However, the conventional manual classification of crimes based on textual descriptions is laborious, prone to human error, and frequently inconsistent. (Eldho,2023)

Moreover, the manual annotation of some visual content can be problematic from an occupational health and safety perspective, as the potential exists for individuals to be exposed to a large number of indecent and disturbing images. To further compound matters, forensic practitioners generally have limited background knowledge of big data, Artificial Intelligence, and data sciences to help create models tailored for their own practices. In order to make full use of large forensic image databases and remove associated practical issues, specialized high-performance and accurate computer vision (e.g., automatic recognition and content filtering) and data analytics tools are required.(Taha, 2024)

Clustering is also another common technique used to predict patterns of geographic criminal history and behavior. Remond and Baveja proposed a case-based reasoning (CBR) system that filtered out police cases for better prediction. It worked on noisy data and explained how police reports show only individual criminal activities and do not address gang crimes. Social networks are also used as a potential source of criminal activity indicators. Sadhana and Sangareddy used Twitter information to find real-time criminal offences. They also used the same data set to find the concentration of crime occurrences and find large-scale hotspots. Thus, several techniques are used for analyzing crimes(Hossain et al., 2020)

From high-level categories to lower-level subtypes, the conventional classification of crime types is based on a hierarchical framework. This tree-based classification lacks tricategory semantic linkages since it recognizes different types of crimes as mutually independent when they do not branch from the same higher-level category. In any event, the prevailing trend is on the initial classification of crime types offered by each nation or local organization, which consists of hierarchical structures from a single point of view, ranging from high-level categories to lower-level subtypes. When crime types are not mutually independent, this conventional tree-based classification recognizes them as such. branch from the same higher-level category; hence, there are no semantic connections across categories(Crivellari & Ristea, 2021)

Police in the United Kingdom use a hierarchical framework of UK Home Office codes, classes, sub-classes, and offence categories to categorize crime events. These classifications correspond to the Notifiable Offence List, which, for each class, corresponds to a pertinent law that is used to prosecute people who are thought to have committed a specific crime. For example, code 28/3, class "Burglary," subclass "Residential Burglary," offence category "28F Attempted Burglary— Residential," and its application related to Section 9 of the Theft Act (1968) are used to record an attempted residential burglary. The aforementioned categories were created to facilitate a variety of criminal justice procedures and allow for comparisons in crime statistics both within and between jurisdictions(Birks et al., 2020)

Using text mining, natural language processing, and machine learning approaches, our system tackles the problem of classifying unstructured legal documents by organizing and analyzing this data and extracting reports related to crime scenes. Our work stands out because we created a crime lexicon with 151 related terms and 70 crime instruments that were taken from different forensic sources. Despite being regarded as rather modest, it helped provide positive categorization results. To enhance crime scene investigation, the suggested crime dictionary can be expanded, used in sophisticated searches, and linked with police databases(Bifari et al., 2024)

For a limited portion of the crime data sets, K-Nearest Neighbor (KNN), Support Vector Machines (SVM), decision trees, random forests, Naive Bayes, and so forth are comparatively more effective. The fundamental framework of crime prediction is shown in Fig. 1. A plan based on conventional machine learning is presented in Part (a). First, a training set and a test set are created from the data set. In order to achieve an appropriate representation of the crime data, the algorithm's fundamental requirement is to extract the crucial information that can represent the crime event data. Since there are many kinds of crime data, it is vital to choose pertinent attribute data and then extract all of the data's features at once to increase the data's capacity for effective expression. The crime is then forecasted using a range of machine learning algorithms, including support vector machines, decision trees, and linear regression models(Yin, 2022)

Researchers have heavily relied on parametric statistical models (e.g., generalized linear models) to predict a variety of criminological outcomes, including drug use, arrest, recidivism, pretrial detention, incarceration, parole release, and inmate misconduct. These models were inspired by actuarial risk assessment tools in econometrics and public health. Their level of predictive accuracy is unimpressive, despite the general agreement that more objective and scientific applications of these statistical techniques perform better than clinical or professional judgments made solely by subjective intuitions with prior experiences or opinions of the decision-makers. For instance, Farrington & Tarling (1985) and Farrington (1987) discovered that the prediction accuracy of commonly used statistical techniques is low, with a percentage of false positives and false negatives higher than 0.5 (Na et al., 2021

2.4 Cybercrime detection in social media

Cybercrime, to put it simply, is any criminal offense that uses a computer, digital devices, or the internet as a conduit. It falls under the category of contemporary crimes brought about by developments in the Internet and related technology. They might be categorized as hybrid offenses because, in contrast to more conventional crimes like robbery and theft, which can be clearly localized in terms of time and place of occurrence, these crimes may have a connection that is challenging to pinpoint. Given the aforementioned, some authors have characterized it as a composite crime that includes cyber-dependent crimes made possible by the Internet and digital technology, as well as cyber-enabled crimes, including organized crime, fraud, forgery, theft, and money laundering. (Chinedu et al., 2021)

Social media is being used for a variety of activities, but it can also lead to cybercrime. Online sharing of personal information can lead to crime. Victims may be reluctant to report crimes due to triviality, embarrassment, or lack of awareness. Social media monitoring can be used to augment conventional crime reporting. Social media is being used to facilitate criminal activity, much like other emerging technologies and communication channels. It is imperative to protect private data being transferred via networks. (Abdalrdha et al., 2023)

As with all cybersecurity and cybercrime issues, awareness is critical to help combat cybercrime on social media. Employees and organizations risk falling victim to cybercrime on social media. Awareness is also required to avert inadvertent perpetration and participation in cybercrime on social media. Organizations must incorporate cybercrime on social media awareness into general cybersecurity and cybercrime awareness training. Organizations must also measure their employees' cybercrime awareness levels on social media to gauge the effectiveness of their awareness training programs. Few studies exist on cybercrime awareness on social media, and most cybercrime awareness measurements do not consider cybercrime on social media.(Umeugo, 2023)

According to the National White Collar Crime Center's "Criminal Use of Social Media" white paper, social media's recent rise has significantly altered the communication environment. Social media sites like Instagram, Facebook, LinkedIn, YouTube, and Twitter are used daily by millions of individuals. People interact with each other via various sites. These channels are used for advertising, hiring new employees, and routine communication within the public sector. (Kaur et al., 2024)

Online social networks (OSNs) such as Twitter are being used more and more in today's society to spread information globally. Online social networks have gained popularity because of their broad reach, streamlined dialogues, anonymity, and capacity to deliver brief bursts of information. Additionally, it could go into a channel for different forms of online harassment and bullying. Common conversations and banter on social networking sites can occasionally turn explosive, leading to the unchecked expression of hate, harassment, and aggressiveness without the possibility of personal danger or reciprocal physical harm. Reports of mental and psychological health problems in recent years have been directly connected to cyberbullying, abuse, and harassment(Osho et al., 2020)

The proliferation of digital communication platforms has led to the emergence of cyberbullying as a major social issue. In contrast to conventional bullying, cyberbullying enables offenders to reach victims at any time and from any place, transcending physical boundaries. Particularly common among teenagers and young adults, this type of harassment has been connected to a number of detrimental effects, such as anxiety, sadness, and even suicidal ideation(Altayeva et al., 2024)

The findings show that many students share too much information through emails and social media, which may pose a threat to them. Most organizations have implemented multilayered security mechanisms, but information security may leak on social sites. Cyberbullying has spread widely, especially among those who are prone to the practice of cyberbullying. Security attacks such as hacking, spoofing, spamming, cyber stalking, Trojans, worms, viruses, and exposure scams are common through social media, hence the ICT policy and government policy have addressed such issues. Higher learning institutions need to invest more resources in training students, awareness, research and development, IT security, and social media crimes. (Benard et al., 2021)

Threats from cybercrime are becoming more frequent every day. Due to a lack of record-keeping at relevant offices for a variety of reasons, including victims' presumptions about police response, users' ignorance of IT (information technology) acts on cybercrimes, and victims' incapacity to identify that they have been victimized, there is currently no foolproof, systematic, and trustworthy tool for reviewing cybercrimes. Reducing and categorizing cybercrimes may also be aided by promoting and educating the public about them, as well as by identifying and maintaining area-specific cybercrime rates. (Ch et al., 2020)

In this regard, we propose a machine learning-based cyberbullying detection model that can identify whether or not a communication is related to cyberbullying. In the suggested cyberbullying detection model, we have looked into a number of machine learning methods, such as Naive Bayes, Vector Machines for Support, Decision Tree, and Random Forest. We use two datasets from posts and comments on Facebook and Twitter for our studies. BoW and TF-IDF are two distinct feature vectors that we employ for performance evaluations. The findings show that the TF-IDF feature outperforms BoW in terms of accuracy, whereas SVM outperforms all other machine learning algorithms employed in their study. (Islam et al., 2020)

In this regard, the current paper suggests using NLP to detect cybercrimes on social media. It used an NLP similarity model to find groupings of social media user accounts that produced messages endorsing hatred and violence, thereby affecting public safety. This final scenario is an unwanted use of ICT that goes beyond acceptable social protest and is seen as a cybercrime since it might be a component of a series of coordinated actions meant to incite instability. The term "campaign of HSM" refers to instability caused by a threat agent, and it is one of the most challenging cyber operations to uncover because it may encounter the same difficulties as any global cyber incident (such as lack of sovereignty, anonymity, and lack of regulation, among others), making it challenging to identify the real threat agent behind such a campaign(Ramírez Sánchez et al., 2021)

In addition to helping to accomplish a number of business objectives, the study of this data has resulted in the development of new technologies and solutions. For instance, social media data has previously been examined for a number of reasons, such as forecasting stock market movements, box office outcomes, and election outcomes. Previous research on Twitter data indicates that people have begun to use the platform to report and communicate a variety of crimes. In Example 1, the user uploaded details on a fictitious Facebook ID and asked the authorities to detain the individual responsible for its creation. This type of cybercrime is documented on Twitter(Lal et al., 2020)

2.5 Challenges in the Classification of Crime-Related Text

Developing highly effective classifiers for such imbalanced datasets remains a significant challenge. As the dataset imbalance increases, the model tends to overfit the majority class, impairing its ability to achieve the desired performance on minority classes and leading to biased outcomes. Hence, the reliability of the model in real-world applications deteriorates, resulting in failures in critical scenarios where the accurate detection of minority classes is essential. (Taskiran et al., 2025)

Multi-label text classification involves assigning more than one class to a text based on analyzing the terms in the text body. presents an open-world multi-label text classification. In this type of task, the possible classes are not available a priori. The method needs to perform topic modeling for subsequent text classification. So, it fragments the texts and sends them to an LLM for keyword extraction. The words are grouped, and the centroids of each group form the labels for text classification. The labels for the text classification procedure can all be available a priori. However, some researchers investigate the need for a dynamic update of the possible labels available for classification.(Vieira et al., 2025)

Text classification can be done manually, but it is time-consuming and has a high cost. Manual classification comes with lots of errors and is less accurate because of human error and a lack of domain knowledge. There was a huge revolution in text classification when machine learning models, such as the Support Vector Machines (SVM), Naive Bayes (NB), and Random Forest (RF) started to replace manual work, because these models not only reduce the time and cost but also are highly accurate for classification. Since the inception of machine learning models, numerous studies have been conducted to enhance, optimize, and refine the text classification process.(Palanivinayagam et al., 2023)

The entire legal text is then divided into several predefined logical segments using structural characteristics. Next, the corresponding entities are extracted from each logical segment using a hybrid method of rules and deep learning. For simple entities, the rules and patterns are generated in the process of knowledge modeling to extract entities. For complex entities, the pre-trained model Bidirectional Encoder Representations from Transformers (BERT) is introduced into the task of legal text information extraction, and is combined with Bi-directional Long Short-Term Memory (Bi-LSTM) and Conditional Random Field (CRF) algorithms to extract entities. It is effective in solving the problem of polysemy in legal texts(Ren et al., 2022)

Beyond the analyses of free-text recorded by police, several studies have also explored how LDA topic models might be used to analyze crime-related phenomena discussed in unstructured content posted online. Combine LDA and collaborative representation classifiers in an attempt to detect ‘criminal intention’ in free-text extracted from online blogs. Relatedly, applies LDA topic models to geotagged tweets posted to the online platform Twitter in Chicago to generate neighborhood dominant topics indicative of the ecology of a particular location. These classifications are then used to positively augment a kernel density estimation-based crime prediction algorithm that seeks to estimate the levels of distinct offences in the immediate future.(Birks et al., 2020)

A large language model (LLM) is a class of artificial intelligence models, typically based on deep learning, specifically designed to comprehend and generate human language. These models are characterized by their enormous size, often containing hundreds of millions or even billions of parameters, which allows them to perform a wide range of natural language understanding and text generation tasks. For crime prediction tasks, an ML model can be trained on inputs to output labels from a fixed set of classes(Sarzaeim et al., 2025)

Securing large language models (LLMs) and NLP technology in general has not been a priority until recently. Yet mass adoption of this technology has led to deployment in contexts where security failures present a risk to individuals, organizations, and society at large—demonstrated by, inter alia, LLM-assisted identity fraud phishing campaigns and automated influence operations If the latest NLP technologies are to withstand an increasing barrage of threats, NLP practitioners must now educate themselves on cybersecurity and cybersecurity practitioners must educate themselves on NLP. (Lent et al., 2025)

As another example, if there is a spike in traffic accidents during a festival period, local police could investigate the causes and allocate appropriate healthcare resources to the affected area. If drunken driving is a significant cause, police could dispatch corresponding real-world statistics provided by administrative authorities. It conducted a case study on the efficacy of the news classification models using two reputable online news sources and seven prevalent types of crimes/accidents in Thailand(Tuarob et al., 2025)

Additionally, the issue of creating software tools and methods to aid in the investigation and prevention of criminal acts based on textual material disseminated online is still unsolved today. Numerous applications are available to handle the problem, such as hate voice recognition, crime prediction, criminal pattern modeling, identification of crime-related topics, etc. However, more effective techniques for analyzing criminal content in computer-mediated communication (CMC) are still required in order to prevent and solve crimes. Crime-related event extraction (CREE) is one of the most difficult and effective methods for identifying and forecasting criminal activity(Khairova et al., 2023)

Over the course of 70 days, a sizable amount of tweets were gathered from the 18 biggest Spanish-speaking nations in Latin America. After classifying these tweets as either crime-related or not, more data is retrieved, such as the type of offense and, if feasible, any city-level geolocation. Approximately 15 out of every 1000 tweets contain content about a crime or fear of crime, according to data analysis. The number of homicides, the murder rate, or the degree of fear of crime as reported in surveys are then compared to the frequency of tweets about crime. The findings indicate that social media has a significant bias in favor of violent or sexual crimes, just like mass media like newspapers. Additionally, there is little correlation between social media messages and criminal activity. (Prieto Curiel et al., 2020)

Unfortunately, there are many methodological and conceptual problems with correctional classification. Many of these problems appear to come from the lack of a methodical procedure or guiding framework for creating classification systems. There is a lack of understanding regarding the task or tasks that a particular classification system can direct, and classification systems are frequently used to achieve several, frequently contradictory objectives. The implication is that we frequently use categories to accomplish tasks that they are not meant to do. Depending on the objectives that the stakeholder is attempting to accomplish, phenomena can be categorized in many ways(Carter et al., 2021)

2.6 Related Work

In recent years, there has been a lot of interest in using artificial intelligence (AI) and machine learning (ML) to analyze crime (Hariguna & Ruangkanjanases, 2023). created a model for an adaptive decision-support system that is specifically for automatically analyzing and sorting crime reports in the context of e-government. Their study built an automated framework that used Natural Language Processing (NLP) techniques to pull out important information from reports that came in. The system was found to greatly improve the efficiency of managing crime data, allowing authorities to respond to crime events more quickly and accurately. Looking at things together (Mussiraliyeva & Baispay, 2024) tested different machine learning methods - old styles and newer deep learning setups - to sort through text about crimes. Rather than making assumptions, they turned to reliable methods - such as the Area Under the Receiver Operating Characteristic Curve

(AUCROC) - to assess which model performed best. This shift led to tangible insights for police departments, revealing patterns in offender behavior along with hints about future actions. These findings didn't just show new tools - they helped shape smarter ways to stop crimes before they grow. Looking closely at tough investigation issues. (Park et al., 2024) created a mixed system to help South Korean law enforcement deal with new pressures after changes to the criminal procedure law in 2021 made cases harder to solve. Instead of relying on manual review, their approach used two existing transformer-based models that had already been trained on legal data - tweaked slightly through additional learning - to pull out 18 crucial details from paperwork automatically. Looking back, it became clear how vital AI tools are when investigators face growing workloads and longer report prep intervals. When it comes to sorting public crime data. (Ali et al.2023)suggested using a BERT-driven system to boost how news documents about crimes are sorted. Because clear information and swift deployment matter in crime prevention, their work highlighted the need for precise data pulling by different user groups. It turned out that using existing language models led to much stronger narrative handling - especially for stories tied to criminal activity - than older approaches did. Focusing first on materials not in English (Carnaz et al., 2021)faced little data by building a labeled set of crime-linked Portuguese texts. Their effort showed how complex crime exams truly are - blending fields like digital enforcement, interview handling, and keeping evidence trails intact. This dataset came together to build a base resource - key for creating NLP and machine learning tools that handle intricate crime cases across languages beyond English. Looking into better training data(Liu et al.2023) explored ways to boost text variety while keeping results accurate. Instead of just adding more samples, they tried swapping words with synonyms, tossing in random terms, or reshuffling full sentences. These shifts helped spread out the examples used to teach models. Because of this mix, the system learned to recognize patterns across deeper layers of meaning in crime classification. Over time, their findings pointed to stronger performance when faced with complex structures within criminal taxonomy. Starting with the challenge of uneven data spread, Franklin et al. (2024)introduced a new approach - "text-guided mix-up" - aimed at handling the longtail issue during sorting jobs. Drawing on how classes are linked in meaning, thanks to insights from pre-trained vision-language tools such as CLI P, they tapped into written guidance to ease imbalances shaped by sparse example sets. Looking at things this way fits well when studying crime patterns - some reported incidents happen much less often but still matter just as much in understanding what's happening. Looking at how African languages work(Sani et al.2025)

Author & Year	Description	Methods	Limitations
Hariguna & Ruangkanjanases (2023)	Developed an adaptive decision support system for e-government crime analysis.	Automated NLP framework to extract key info from incoming reports.	Traditional ML methods fail to capture semantic and contextual nuances.
Mussiraliyeva & Baispay (2024)	Evaluated effectiveness in detecting and categorizing crimerelated text.	Comparative analysis of traditional vs. deep learning models.	Challenges in harnessing the vast amount of online textual data.
Park et al. (2024)	AI system to help South Korean police handle increased workloads.	Hybrid model finetuning two Transformer-based pretrained models.	Extended time required for manual report preparation remains a burden.
Ali et al. (2023)	Aimed to strengthen fighting capacities against crime through data extraction.	BERT-driven system to boost news document sorting.	Exploding volume of crime-related data makes manual possession difficult.

Carnaz et al. (2021)	Focused on non-English (Portuguese) crimelinked text classification.	Construction of a labeled dataset for forensic tasks.	Scarcity of labeled data in languages other than English.
Liu et al. (2023)	Explored data diversity in hierarchical multilabel classification.	Methods: Synonym replacement, random insertion/deletion/swap.	Excessive data enhancement may lead to model over-fitting.
Franklin et al. (2024)	Addressed longtailed label distribution in training.	Text-guided mix-up technique using semantic relations.	Traditional deep learning requires costly, large, balanced datasets.
Sani et al. (2025)	Investigated sentiment analysis	Language-Adaptive Fine-Tuning (LAFT) using Afri-BERTa.	Smaller languages often just copy English-based

2.7 Proposed System

The proposed system is an NLP (Natural Language Processing) based text classification system that classifies any given social media post as crime-related or non-crime-related. The dataset used for this proposed system consists of social media posts, which are manually labelled by humans as crime-related or non-crime-related. The output is generated after the preprocessing of the input text and then passing it to the machine learning and deep learning algorithms. The algorithms used in this proposed system are Support Vector Machine (SVM), Random Forest, and BERT. The preprocessed features of the input text are used by these algorithms to classify and find out whether the input text is crime-related or not. The proposed system is trained and tested using a manually labelled dataset, and the performance is evaluated by calculating the accuracy, precision, recall, and F1-score. The proposed system has a simple web interface developed in Python and Flask. The interface allows users to input any social media post that needs to be classified. The proposed system then classifies and provides the output of whether the input social media post is crime-related or not, along with the confidence. The proposed system thus facilitates an efficient monitoring of the social media posts as compared to the manual surveillance required in traditional systems.

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